

The TerraSAR-X Satellite

Wolfgang Pitz and David Miller

Abstract—TerraSAR-X is a versatile synthetic aperture radar (SAR) satellite with active phased array antenna technology and represents the backbone of the German national radar Earth observation mission. With its large variety of different SAR imaging modes and its high operational flexibility, TerraSAR-X ideally serves the scientific community and users from the industrial sector and governmental institutions. The innovative satellite system design combines the rich experience from past German and European SAR space missions like X-SAR, SRTM, ERS 1 and 2, and Envisat combined with state-of-the-art Earth observation bus technology as used, e.g., on the CHAMP and GRACE satellites.

Index Terms—SAR satellite, synthetic aperture radar (SAR).

I. INTRODUCTION

REMOTE sensing of the Earth's surface from space by an active microwave instrument allows observation independent of time of day or cloud cover. The synthetic aperture radar (SAR) principle and development of key technologies in space quality have allowed to steadily improve geometric resolution and radiometry sensitivity.

European spaceborne SAR programs started in the 1980s with the ERS-1 satellite carrying a passive C-band antenna. The X-SAR instrument as the German contribution to the National Aeronautics and Space Administration's Space Shuttle missions SIR-C/X-SAR and the Shuttle Radar Topographic Mission (SRTM) followed. Improved operational flexibility, increased image resolution, and the topographic information content provided by this first spaceborne interferometer impressively demonstrated the capabilities of X-band spaceborne radar. This experiment established the basis for a successful German X-band SAR program to develop key technologies for modern phased array antennas. The technology demonstrator of an active phased array SAR antenna (DESA), developed in the early 1990s, became the key element of the active antenna of TerraSAR-X (Fig. 1).

The TerraSAR-X satellite is a further step on the path of increasing performance and also provides a unique flexible service to users, allowing the development of new SAR methods and a wide range of applications. The TerraSAR-X mission provides 2-D Earth surface imaging with geometric resolution down to 1 m and noise equivalent sigma zero below -19 dB. In a formation flight together with its future partner TanDEM-X,

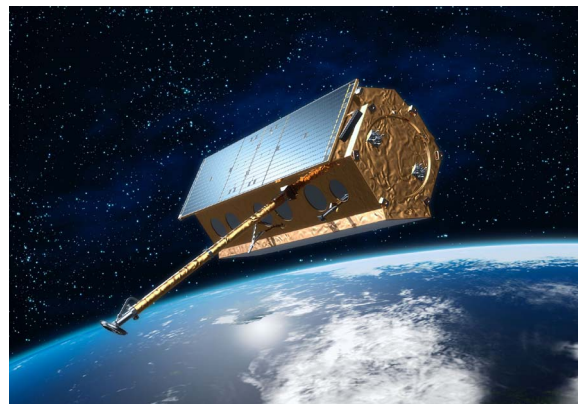


Fig. 1. TerraSAR-X S/C.

3-D imaging of the complete landmass of the Earth will be performed with a height resolution of 2 m.

This paper treats the design and performance of the bus and the SAR (Section II), the additional payloads (Section III), and satellite prelaunch testing (Section IV). Section V describes the current in-orbit status of TerraSAR-X. Then, Section VI gives an outlook on the TanDEM-X satellite and mission.

II. DESIGN AND PERFORMANCE

A. Space Segment

The technological foundations of the TerraSAR-X space segment design were laid in the final decade of the last millennium with the successful Space Shuttle missions X-SAR [1] and SRTM [2], [3] regarding the radar instrument and with the low Earth orbit (LEO) science missions CHAMP [4] and GRACE [5] for the satellite bus.

Furthermore, the German Aerospace Center (DLR) funded technology studies which played an indispensable role in the reduction of risk connected with the introduction of new SAR technologies applied on TerraSAR-X. Of key importance was the DESA (X-band SAR Antenna Demonstrator) study, where a prequalification model of one of the 12 TerraSAR-X antenna panels was subjected to rigorous functional and performance testing, and the TOPAS (Technology Development for Onboard SAR Processing and Storage Demonstrator) study, where novel SAR data onboard processing and storing techniques were investigated and prequalified. On the basis of this rich heritage, it was possible to realize an efficient design which offers an excellent performance-to-cost ratio.

The TerraSAR-X space segment was optimized for operation in LEO in keeping with the required radar performance and technology capability (the mission orbit characteristics are listed in Table I) and is compatible with a wide range of launchers due to its compact shape and low mass.

Manuscript received March 18, 2009; revised July 21, 2009 and October 19, 2009. Current version published January 20, 2010.

The authors are with Astrium Satellites, 88039 Friedrichshafen, Germany (e-mail: wolfgang.pitz@astrium.eads.net; david.miller@astrium.eads.net).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TGRS.2009.2037432

TABLE I
TerraSAR-X MISSION ORBIT

Orbit Parameter	Value
Orbit Type	Sun-synchronous
Repeat Period	11 days / 167 orbits
Orbits per day	15 2/11
Altitude at equator	514 km
Inclination	97.44°
Local time at Ascending Node	18:00 hours

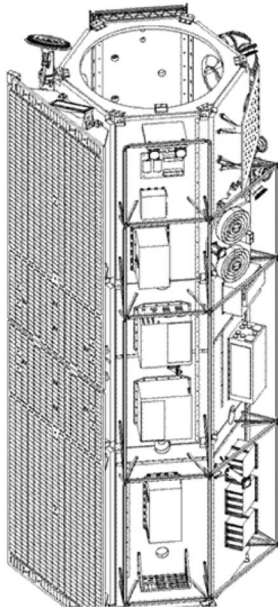


Fig. 2. Zenith side of the S/C showing the solar array.

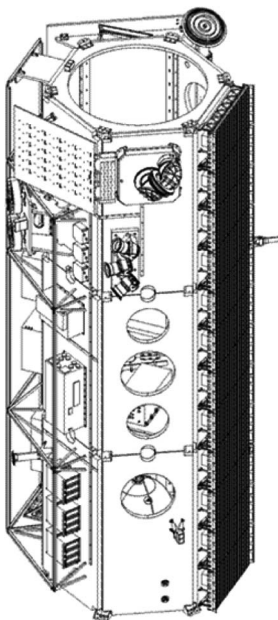


Fig. 3. Right side of the S/C.

The hexagonal outer shape of the spacecraft (S/C) efficiently fits the shape of a typical launcher fairing. The body-mounted solar array is seen on the left in Fig. 2. The radar antenna is seen on the right in Fig. 3. The majority of the electronic units are

accommodated on the outside of the basic structural hexagon, which allows easy access during integration and test. The main dimensions are 4880 mm in length and 2400-mm maximum width. In nominal flight attitude, the SAR antenna off-nadir angle amounts to 33.8°. The extensive use of lightweight carbon fiber-reinforced plastic (CFRP) technology in a satellite structure and a SAR antenna leads to a low overall wet mass (i.e., including fuel) of about 1350 kg, including additional payloads.

All systems on TerraSAR-X are single failure tolerant, and the reliability figure (i.e., predicted probability of maintaining the required functionality/performance) is 0.8 after 5.5 years of nominal mission lifetime.

B. Satellite Bus

The main system performance critical tasks of the TerraSAR-X satellite bus are to provide the SAR instrument with sufficient energy and precise pointing and also to maintain a stable orbit and adequate thermal conditions.

The onboard computer includes a processor module based on the ERC32-MCM (the highly integrated multichip-module development of the radiation-tolerant 32-bit SPARC V7 micro-processor) which provides > 10 million instructions/s computing speed.

Electrical power is generated by means of a 5.25-m² large GaAs triple-junction solar array, stored in a 108-Ah (begin-of-life) Li-ion battery and distributed over a 50-V dc main bus system. Under worst case seasonal and end-of-life circumstances and assuming a solar array string failure, the available power is still sufficient for a minimum SAR instrument operation of 170 s/orbit in average, with 10% margin. However, apart from a small region around summer solstice, where eclipses with a maximum duration of 22.4 min are experienced, electrical energy is abundant so that SAR imaging of up to 10 min/orbit is typically possible.

High-precision satellite pointing control and determination are achieved on the basis of star trackers located close to the SAR antenna plane so that an antenna boresight pointing accuracy of 65 arcsec (3σ) is reached. Nominal attitude control follows a novel total zero Doppler steering law developed by DLR [6]. Precise orbit determination (POD) is performed with a two-frequency GPS receiver and raw data postprocessing on ground, which allows for orbit restitution accuracies in the centimeter range [7]. A set of high-torque reaction wheels enables rapid rotation into the so-called sun side-looking (SSL) orientation (whereby the SAR looks to the left side of the subtrack) which is used to improve access of high-priority imaging targets. In order to point the SAR antenna into the SSL direction, a roll movement of 67.6° is necessary, which is achieved in less than 180 s.

The satellite bus is equipped with a monopropellant hydrazine blowdown-mode propulsion system for orbit maintenance and safe-mode attitude control. A total of 78-kg propellant loading, providing a delta v (total thrust) of 105 m/s, is sufficient for almost ten years of operation, enabling the satellite to be kept within a ± 250 -m orbit tube during the entire life to allow for repeat-pass SAR interferometry. The attitude and orbit control system is completed by sensors to

TABLE II
BASIC SAR MODE PERFORMANCE

	High-Resolution Spotlight Mode	StripMap Mode	ScanSAR Mode
Ground Resolution			
- Across Track	1 m	3 m	16 m
- Along Track	1 m	3 m	16 m
Product Coverage			
- Along	5 km	<1500 km	<1500 km
- Across	10 km	30 km	100 km
Access Range (at equator)			
- Full Performance	20°-55° (463 km)	20°-45° (287 km)	20°-45° (287 km)
- Data Collection	15°-60° (622 km)	15°-60° (622 km)	20°-60° (577 km)
NESZ			
- typical	-20 dB	-22 dB	-21 dB
- worst case	-16 dB	-19 dB	-19 dB
Data Rate			
- mean, 8:4 BAQ	680 Mbps	580 Mbps	580 Mbps

determine attitude (coarse Earth and sun sensor) and attitude rates (inertial measurement unit and magnetometer) and also a magnet-torquer actuator for attitude control.

Regarding thermal control, a combination of multilayer insulation, radiator foils, and software-controlled heaters guarantees a low-variation temperature environment over all seasons, satellite modes, and orientations. All performance-sensitive radar elements accommodated within the satellite bus are closely thermally conditioned using foil heaters within a range of less than 1 K.

Command and control of the satellite are performed in an encrypted way supported by an S-band telemetry, tracking, and command system at data rates of 4 kb/s for uplink, 32 kb/s for real-time housekeeping (HK) downlink, and 1 Mb/s for HK dump downlink. SAR data are transmitted to the ground with an X-band system at a net data rate of 270 Mb/s. The X-band downlink antenna is mounted on a 3.3-m-long deployable boom—the only mechanism on the satellite—in order to prevent interference with the X-band SAR instrument. This concept enables for simultaneous SAR data acquisition and data downlink.

The bus units are generally redundant to provide tolerance to any single failure. Some hardware is “hot” redundant (i.e., constantly activated) for fast reaction or other reasons.

C. SAR Instrument

The key characteristic feature of TerraSAR-X is the advanced high-resolution X-band SAR based on active phased array technology. The specified performance parameters for each of the three basic SAR modes, namely, spotlight, stripmap, and ScanSAR, are presented in Table II.

1) *SAR Antenna*: The SAR center frequency is 9.65 GHz, and the bandwidth is selectable up to a maximum of 300 MHz. The SAR antenna is capable of operation in two polarizations, namely, H and V, and consists of 12 panels each with 32 slotted waveguide subarrays. Every subarray is equipped with

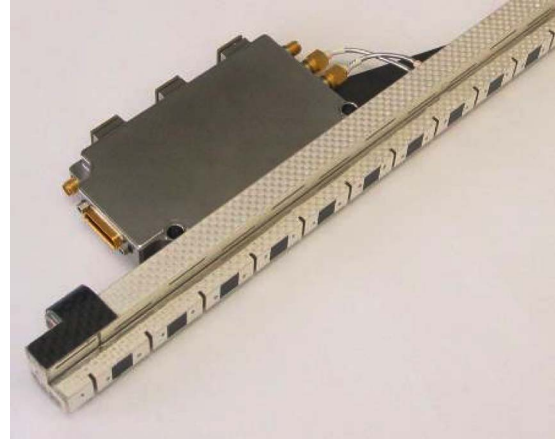


Fig. 4. TerraSAR-X subarray with TRM.

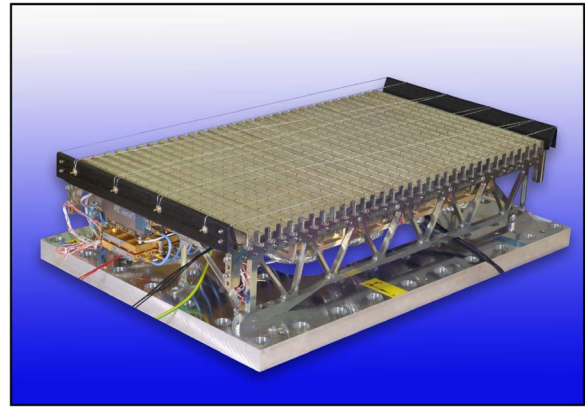


Fig. 5. TerraSAR-X antenna panel.

a transmit/receive module (TRM), which totals to 384 TRMs for the complete SAR antenna. Fig. 4 shows a view of a single active subarray (glued pair of H and V slotted waveguides with its TRM). Fig. 5 shows an integrated antenna panel (array of 32 active subarrays with signal distribution networks and power conditioners).

The approximate dimensions of the antenna are 4800 mm in length, 800 mm in width, and 150 mm in depth. The beam steering capability is $\pm 0.75^\circ$ in azimuth and $\pm 20^\circ$ in elevation. A highly efficient thermal conditioning system guarantees low temperature gradients over the antenna surface and a low temporal heat-up gradient. Due to this feature, continuous imaging times of up to 10 min are possible without overheating the antenna. Advanced CFRP technology is used for the subarrays, which almost completely prevents thermomechanical distortions on the antenna. (See [8] for details.) The TRMs are powered by an ac bus system which up converts the 50-V dc of the main bus into 115 V/30 kHz ac. The ac voltage is then distributed to each antenna panel, where, on panel level, it is down converted and rectified to supply the TRMs with the necessary voltages.

2) *Instrument Units*: The antenna beam is determined by the central antenna control electronics unit which provides each TRM via a data bus system with amplitude, phase, and polarization setting information for each radar pulse. A large variety of pulse forms and bandwidths can be created due to the freely programmable digital chirp generator. An extensive

calibration network covering virtually all radio-frequency (RF) paths within the instrument up to the TRMs serves as essential basis for in-flight signal calibration. The SAR data generated by the instrument can be compressed following an online block adaptive quantizer scheme with selectable ratios of 8:6, 8:4, 8:3, or 8:2 before they are stored in the solid-state mass memory unit with 384-Gb begin-of-life and 256-Gb end-of-life capacity. Receive, transmit, and calibration signal paths of the radar electronics and the antenna are cross-strapped by means of a redundancy network. The instrument electronics and redundancy network layout also allows for operation in the dual-receive antenna mode, where the echoes received by individual azimuth antenna halves can be separated during ground processing. This facilitates along-track interferometry and full-polarimetric data acquisition [9], [10].

3) *Internal Calibration:* The SAR design includes an internal calibration scheme whereby transmit signals are routed into the receiver to allow monitoring of amplitude/phase during operation (either during on-ground testing or in orbit). This scheme produces data which can be used by ground processing for radiometric and/or phase correction of the image data.

The internal calibration route is not exactly the same as the nominal radar transmit/receive path. On the one hand, some elements of the transmit/receive path are not fully covered, e.g., the internal calibration signals are coupled out at the front-end TRMs (transmit signal toward subarrays), hence undergo a different path than the radar signal itself. On the other hand, some elements of the internal calibration path are not part of the nominal radar path, e.g., the aforementioned internal calibration signals are routed back to the RF electronics via dedicated coaxial cables.

Temperature variations of those elements which are not fully covered by or specific to the internal calibration loops result in instabilities which cannot be corrected by using the internal calibration data. On-ground measurements (i.e., characterization) of these elements (viz., amplitude and phase variation with frequency and temperature) allow a second level of correction in the ground processing.

Fig. 6 shows the results of on-ground verification of the internal calibration scheme: blue for power-gain product determined by internal calibration, red for power-gain product determined using an external stimulus, and their difference (in black) for operation using internal calibration every 60 s and deactivation of the TRM automatic temperature compensation in between. (See [11] for details.) The smooth drift allows the accurate interpolation of the internal calibration measurements. This verification proves the performance of the internal calibration scheme with a single antenna panel. This scheme then can be used (without external equipment) for performance verification during thermal-vacuum testing with the complete antenna and, based on this successful verification, during in-orbit operation to ensure radiometric stability.

III. ADDITIONAL PAYLOADS

A. LCT

The Laser Communication Terminal (LCT) is an experimental payload developed and built with funding from

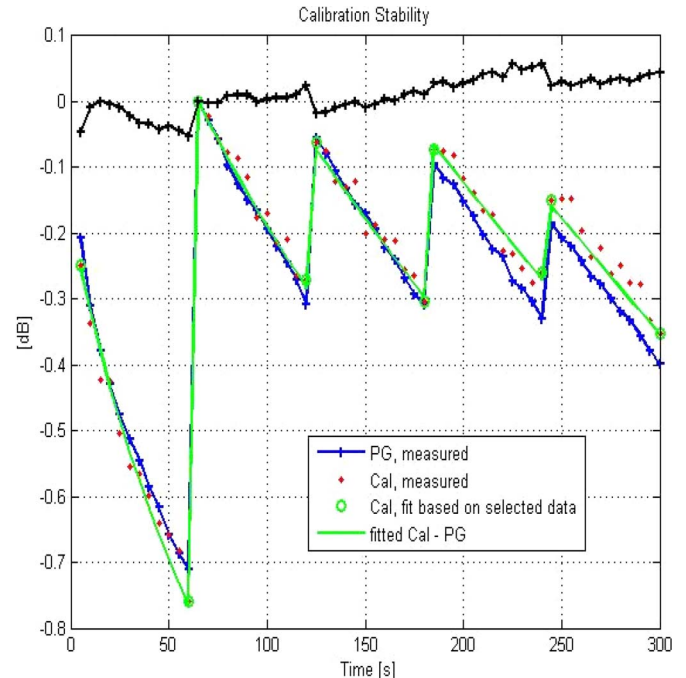


Fig. 6. TerraSAR-X test result for radar amplitude stability.

DLR/Bundesministerium für Bildung und Forschung (BMBF) under leadership of the industrial prime contractor Tesat-Spacecom with major contributions from EADS Astrium and Zeiss Optronik [12]. The main aim of the LCT experiment is to demonstrate the feasibility of laser-based communication between satellites in different orbits at high data rates. For this purpose, the LCT on TerraSAR-X was used in a first verification step to transmit SAR data to a ground-based terminal at a speed of 5.625 Gb/s using a 1064-nm diode-pumped solid-state laser.

In a further verification step, an intersatellite link (ISL) was established with the U.S. Missile Defense Agency's Near Field Infrared Experiment satellite. The LCT was the first laser communication system in space operating on a coherent basis with a homodyne digital receiver and binary phase-shift keying modulation. The optical output power ranges between 0.5 and 1.0 W, and the bit error rate is below 10^{-8} for a distance of up to 8000 km and below 10^{-4} for the link through the atmosphere.

The LCT is accommodated on the antisun side of TerraSAR-X to guarantee a free hemispherical field of view which also includes the Earth surface. The terminal consists of one physical unit with a mass of approximately 33 kg and an average power consumption of 136 W, functionally divided into an optics unit including a coarse pointing mechanism, a telescope, a fine pointer and a receiver, and a frame unit containing the electronics. For thermal control reasons, the LCT is supported by a heat pipe radiator with an area of 0.57 m² and a weight of approximately 10 kg. Due to the abundant TerraSAR-X energy margin, the LCT can be operated simultaneously with SAR imaging during most of the year.

B. TOR Equipment

The Tracking, Occultation, and Ranging (TOR) equipment is furnished by GeoForschungsZentrum, Potsdam, Germany,

TABLE III
VIBRATION TEST LOADS

Frequency in Hz	Axial Loads in g (vertical in Fig. 7)	Lateral Loads in g (horiz in Fig. 7)
2-5		0,16 – 0,4
5 -10	0,4	0,4
10 – 15	0,48	0,4 - 0,8
15 – 20	0,4	0,4
20 – 40	0,4 – 0,6	0,4 - 0,6
40 – 80	0,6 – 0,8	0,6 - 0,8
80 – 100	0,8	0,8
Sweep rate 2 oct/min		

and the Center for Space Research of the University of Texas in Austin, USA, with cofunding from DLR/BMBF. The TOR instrument package consists of the redundant dual-frequency GPS tracking receiver system IGOR (integrated global navigation solution system and occultation receiver) and a laser retro reflector set (LRR) [13]. The IGOR is an enhancement of the BlackJack receiver, incorporating, e.g., additional receive filtering to avoid SAR interference and hi-rel parts where possible. The GPS tracking data acquired by IGOR are the basis for POD processing on ground. Orbit products with accuracies of < 2 m in position, to be used for SAR image processing and geolocation, are made available within a short time frame, which is typically less than 24 h, after reception of IGOR data on ground. A further TOR mission objective is to collect atmospheric and ionospheric radio occultation data with IGOR, which are used for the improvement of weather forecast and for climatic studies. The LRR is a passive optical device enabling for precise laser tracking from ground, supported by the global network of laser tracking stations of the International Laser Ranging Service. The LRR on TerraSAR-X allows for orbit determination accuracies in the centimeter range and has mainly been used to independently verify the quality of the onboard GPS tracking data. The total weight of the TOR package is less than 10 kg, and its power consumption is below 17 W in average.

IV. TERRASAR-X PRELAUNCH SATELLITE TESTING

The TerraSAR-X satellite was subjected to a classical system environmental test campaign of the following (in sequence): 1) mass properties; 2) separation shock; 3) vibration; 4) boom release; 5) thermal balance/thermal vacuum (TB/TV); 6) electromagnetic compatibility; and 7) acoustic noise testing. All tests were performed over a period of about three months in summer 2006 at the Industrieanlagen-Betriebsgesellschaft mbH facility in Munich.

A. Vibration Testing

The vibration test demonstrated the satellite's capability to withstand the dynamic load environment during the launch and verified the stiffness requirements imposed by the launcher. The test program consisted for each of the three major geometrical satellite axes of a low-level sine sweep to determine the first eigenfrequency, followed by an acceptance level run with loads as shown in Table III. It was concluded by a further low-level



Fig. 7. TerraSAR-X vibration test.

sine sweep to verify the integrity of the structure. Fig. 7 shows the satellite on the vibration shaker.

B. TB/TV Testing

The TB/TV test, which included the simulation of solar and Earth radiation, demonstrated the suitability of the thermal design and the proper function of the overall system in vacuum under extreme temperature conditions. The overall duration of the TB/TV test was ten days, subdivided into a hot testing period with typical satellite internal temperatures above 30°C and a cold testing period with typical internal temperatures below -10°C . For most of the test duration, the vacuum quality was better than 10^{-6} mbar. The TB/TV test also incorporated SAR instrument tests with radiating SAR antenna, which were very useful to verify the predicted heat-up behavior of the complete antenna during a typical SAR data take in space. For this purpose, the TV chamber had been equipped with a vacuum proof RF damping material. Fig. 8 shows the satellite inside the vacuum chamber before closure. Earth radiation is simulated by temperature-controlled Earth simulator panels and the thermal space environment by LN_2 -cooled shrouds, placed around the test chamber walls. The sun simulator beam enters the chamber from below, is reflected by a mirror at the far end of the chamber, and illuminates the satellite solar array.

V. TERRASAR-X IN-ORBIT STATUS

The TerraSAR-X S/C was launched on a Dnepr rocket from Baikonur in Kazakhstan on June 15, 2007. The S/C is working nominally, and excellent images are being continuously produced on ground from the downlinked radar data. All redundant hardware is available except for one (of 384) antenna TRM



Fig. 8. TerraSAR-X TB/TV test.

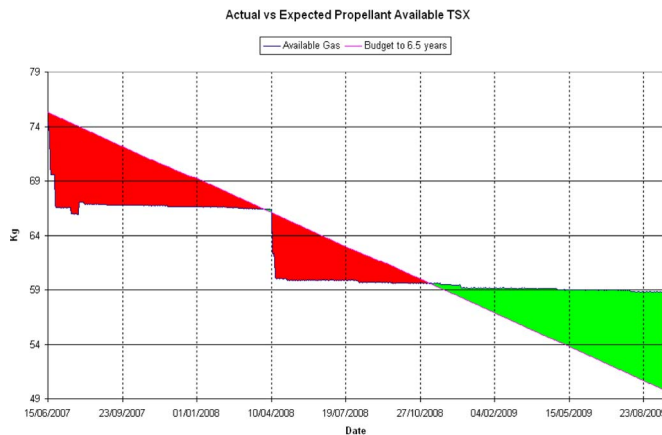


Fig. 9. TerraSAR-X fuel consumption.

which was deactivated before the launch following anomalous telemetry data.

There is a good status of consumables and predictions that show that this fits the planning for the TanDEM-X mission. Fig. 9 shows the fuel consumption since launch compared to an assumed linear decrease over 6.5 years: Green/red indicates a fuel status having more/less than the linear assumption, respectively. A radiation upset in April 2008 did lead to unexpected use of fuel, but a software improvement solved the problem. In the meantime, there is a positive fuel budget.

The condition of the battery is constantly monitored by checking that its voltage during operation does not drop too low and approach the level where an automatic onboard reaction would occur. Every year, a dedicated data take is performed to provide data to allow prediction of the remaining capacity. Fig. 10 shows the results. The solid curve is the original (at launch) prediction; the dashed curve is the updated prediction; today, about 84% of the original capacity is still available.

VI. TANDEM-X OUTLOOK

The TanDEM-X mission became a vision during the development of the TerraSAR-X satellite. The TanDEM-X mission

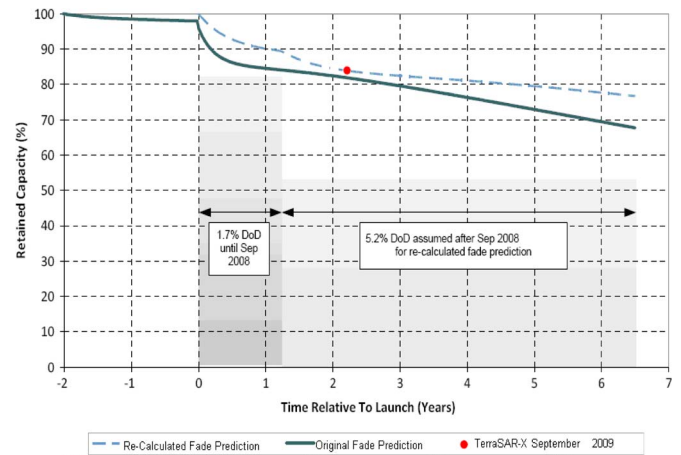


Fig. 10. TerraSAR-X battery capacity.

is the parallel operation of a second satellite in close formation to collect radar interferometric data to derive a global digital elevation model (DEM) of the complete Earth landmass. This needed a minimum extension of the SAR design on TerraSAR-X to support the synchronized operation of both radars. For the bus, the approach was constrained by only allowing software changes on TerraSAR-X. The bus design on TanDEM-X was extended to allow formation flight of both satellites. The software changes are to be verified during the TanDEM-X on-ground tests and uplinked to TerraSAR-X in preparation for the formation flight. TanDEM-X is currently under test, and its launch is planned at the end of 2009 to allow three-year lifetime overlap with TerraSAR-X. For more details, see [14]–[17].

An additional propulsion system based on high-pressure nitrogen gas is accommodated on TanDEM-X. This cold gas system provides smaller impulses than the hydrazine system on both satellites (which is used for orbit maintenance) and supports formation flying by fine orbit control of the TanDEM-X satellite.

The TerraSAR-X instrument was equipped with features required for pulse repetition frequency and phase synchronization between the two satellites. Phase synchronization is facilitated by exchange of radar pulses between both SARs—the so-called “sync” pulses. Six horn antennas on each satellite provide quasi-omnidirectional coverage, as shown in Fig. 11. The red cones symbolize the pointing directions of the respective horn antennas. Their accommodation was heavily constrained by the launcher envelope. Measurements with simulated representative blockage were made to confirm satisfactory antenna field of view.

For DEM data collection, formation flight is necessary with separations down to about 300 m. This infers risk of collision of the satellites and risk of mutual illumination by the main beams of the radar antennas. A number of mechanisms have been introduced on both satellites to safeguard against these risks.

- 1) An additional safe mode using magnet torquers to avoid change in altitude in the case of a radiation upset which could lead to risk of collision.
- 2) An exclusion zone logic to suppress SAR transmission in the case of attempted operation in a forbidden orbit

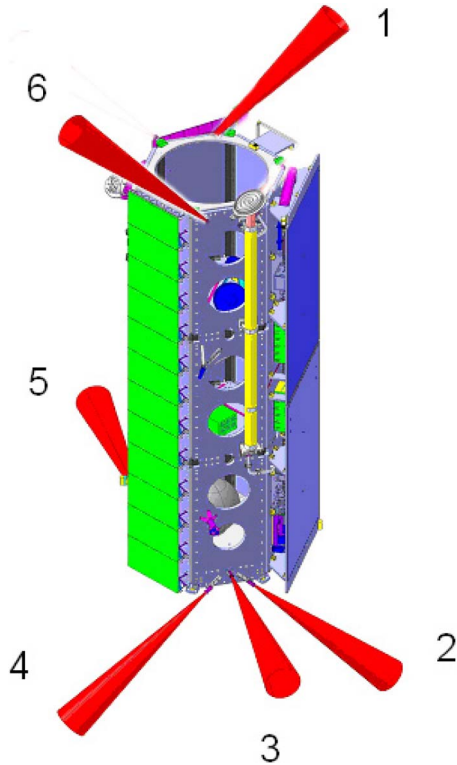


Fig. 11. TerraSAR-X and TanDEM-X sync horns.

segment where receive-only operation is planned. Receive-only operation is used during bistatic operation of the radars for DEM data collection: One radar transmits, and both radars receive.

- 3) A sync warning logic to suppress SAR transmission in the case that low power levels of sync pulses infer a possibly incorrect orbit position of the partner satellite.
- 4) Furthermore, an ISL receiver and a decoder on TanDEM-X allow “listening” to the TerraSAR-X low-rate S-band telemetry. In the case that TerraSAR-X is in safe mode, TanDEM-X suppresses radar transmission considering the concern of radar illumination of TerraSAR-X.

VII. CONCLUSION AND OUTLOOK

The design and performance of the bus, SAR, and additional payloads have been described for the TerraSAR-X and TanDEM-X satellites. This covers the architecture and technology of the SAR active antenna and also the internal calibration scheme to explain how the SAR flexibility and stability are achieved. An outline of the prelaunch test program underlines the approach to secure the functionality and performance in orbit. A summary of the current TerraSAR-X in-orbit health status confirms the design and test approach and provides the best foundation for the success of TanDEM-X. Safeguards for close formation flight have been developed and will be tested in orbit before the begin of this exciting and novel phase.

In the meantime, the next generation SAR instrument has already entered its definition phase. A clear trend toward digital beamforming can be observed, providing high geometric resolution in combination with a very wide swath [18]. To achieve this, new techniques such as, for example, scan on receive or

multiaperture processing are required. As a consequence, the analog-to-digital conversion of the radar echo will be performed on an antenna subaperture level, providing digital input to the subsequent beamforming units.

The demonstration of the feasibility and performance of a digital beamforming antenna is subject of the “High Resolution Wide Swath” SAR Demonstrator that is currently under development at Astrium in Friedrichshafen [19].

REFERENCES

- [1] D. Evans, C. Elachi, B. Holt, M. Kobrick, H. Ottl, P. Pampaloni, M. Schier, E. R. Stofan, M. Vogt, S. Wall, J. van Zyl, and J. B. Way, “The shuttle imaging Radar-C and X-SAR mission,” *Eos, Trans. Amer. Geophys. Union*, vol. 74, no. 13, pp. 145–158, 1993.
- [2] T. G. Farr, P. A. Rosen, E. Caro, R. Crippen, R. Duren, S. Hensley, M. Kobrick, M. Paller, E. Rodriguez, L. Roth, D. Seal, S. Shaffer, J. Shimada, J. Umland, M. Werner, M. Oskin, D. Burbank, and D. Alsdorf, “The Shuttle Radar Topography Mission,” *Rev. Geophys.*, vol. 45, no. 2, p. RG2004, 2007. DOI: 10.1029/2005RG000183.
- [3] A. Nico, M. Eineder, H. Breit, and S. Suchandt, “Shuttle Radar Topography Mission (SRTM): DLR’s interferometric SAR processor for the generation of a global digital elevation model,” in *Proc. 2nd Int. Workshop ERS SAR Interferometry FRINGE*, Liège, Belgium, 1999.
- [4] C. Reigber, H. Luhr, and P. Schwintzer, “CHAMP mission status,” *Adv. Space Res.*, vol. 30, no. 2, pp. 129–134, Jul. 2002.
- [5] B. D. Tapley, S. Bettadpur, M. Watkins, and C. Reigber, “GRACE: The gravity recovery and climate experiment—A mission overview and early results,” *Geophys. Res. Lett.*, vol. 31, no. 9, p. L09607, 2004.
- [6] H. Fiedler, E. Boerner, J. Mittermayer, and G. Krieger, “Total zero Doppler steering—A new method for minimizing the Doppler centroid,” *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 141–145, Apr. 2005.
- [7] Y. T. Yoon, M. Eineder, N. Yague-Martinez, and O. Montenbruck, “TerraSAR-X precise trajectory estimation and quality assessment,” *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1859–1868, Jun. 2009.
- [8] M. Stangl, R. Werninghaus, and R. Zahn, “The TerraSAR-X active phased array antenna,” in *Proc. IEEE Phased Array Conf.*, Boston, MA, 2003, pp. 70–75.
- [9] S. Suchandt, H. Runge, H. Breit, U. Steinbrecher, A. Kotenkov, and U. Bals, “Automatic extraction of traffic flows using TerraSAR-X along-track interferometry,” *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 2, pp. 807–819, Feb. 2010.
- [10] J. Mittermayer, M. Younis, R. Metz, S. Wollstadt, J. Márquez Martínez, and A. Meta, “TerraSAR-X system performance characterization and verification,” *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 2, pp. 660–676, Feb. 2010.
- [11] D. Miller, M. Stangl, and R. Metz, “On-ground testing of TerraSAR-X instrument,” in *Proc. EUSAR Conf.*, Dresden, Germany, May 2006, pp. 16–18.
- [12] B. Smutny, R. Lange, H. Kämpfner, D. Dallmann, G. Mühlwinkel, M. Reinhardt, K. Saucke, U. Sterr, B. Wandernoth, and R. Czichy, “In-orbit verification of optical inter-satellite communication links based on homodyne BPSK,” in *Proc. Publ. Photon. West, LASE*, 2008, p. 687 702.
- [13] M. Rothacher, B. D. Tapley, C. Reigber, R. Koenig, C. Falck, L. Grunwaldt, W. Koehler, F.-H. Massmann, and G. Michalak, “The TOR instrument on TerraSAR-X and TanDEM-X,” in *Proc. IGARSS*, Barcelona, Spain, Jul. 2007, pp. 4983–4986.
- [14] D. Miller, “The TanDEM-X satellite,” in *Proc. EUSAR Conf.*, Friedrichshafen, Germany, Jun. 2–5, 2008.
- [15] M. Bartusch, H. Berg, and O. Siebertz, “The TanDEM-X mission,” in *Proc. EUSAR Conf.*, Friedrichshafen, Germany, Jun. 2–5, 2008, pp. 27–30.
- [16] M. Zink, G. Krieger, H. Fiedler, I. Hajnsek, and A. Moreira, “The TanDEM-X mission concept,” in *Proc. EUSAR Conf.*, Friedrichshafen, Germany, Jun. 2–5, 2008.
- [17] G. Krieger, A. Moreira, H. Fiedler, I. Hajnsek, M. Werner, M. Younis, and M. Zink, “TanDEM-X: A satellite formation for high-resolution SAR interferometry,” *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3317–3341, Nov. 2007.
- [18] G. Krieger, N. Gebert, M. Younis, F. Bordon, A. Patyuchenko, and A. Moreira, “Advanced concepts for ultra-wide swath SAR imaging,” in *Proc. 7th EUSAR*, Friedrichshafen, Germany, Jun. 2008.
- [19] C. Fischer, C. Schaefer, and C. Heer, “Technology development for the HRWS (High Resolution Wide Swath) SAR,” in *Proc. IRS*, Cologne, Germany, Sep. 2007.



Wolfgang Pitz was born in 1961 in Ludwigsburg, Germany. He received the Master of Aerospace Engineering at Stuttgart University, Stuttgart, Germany, in 1988.

He joined the space industry in the same year. Staying with his first employer, formerly Dornier Systems and now Astrium Satellites, he first worked as Thermal Control Engineer on various satellite projects and became Project Manager of the German gravity mission CHAMP in 1996. In parallel to the work on CHAMP he led a series of national and

international satellite early-definition phases, most notably for the DIVA and ROCSAT-2 missions. In 2000 he became Project Manager of the German DLR SAR satellite TerraSAR-X. Starting from 2005 he played a major role in the definition of the German DLR SAR satellite TanDEM-X. After the launch of TerraSAR-X in 2007 he became Project Manager of the SAR instrument for the ESA GMES mission Sentinel-1. Since 2009 he has been Head of Industrialisation of the Astrium Satellites Earth Observation and Science Division.



David Miller was born in 1955 in Liverpool, U.K. He received the B.S. degree in physics from Bristol University, Bristol, U.K.

He joined British Aerospace in 1977 and gained first experience on avionics and radar systems. In 1981, he moved to Germany and joined Dornier Systems which later became Astrium. After performance analysis of the ERS-1 Active Microwave Instrument and the Shuttle X-SAR experiment, he became Project Manager for the Advanced Scatterometer on Metop. He became responsible for the

radar on TerraSAR-X in 2002 and became Manager of the TanDEM-X satellite in 2006.